30 An Open-Source Workflow for Recording the Excavation Process: Volumetric Reconstruction from iPad LiDAR Data at Ca' Nova di Albareto (Parma)

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Digital recording has transformed the documentation of archaeological excavations, yet standard workflows still record only the upper and lower surfaces of each stratigraphic unit, leaving the volume between them, and the stages of its removal, largely undocumented. This gap is particularly relevant because archaeological deposits are three-dimensional volumes, and understanding how they formed requires recording them as such. This paper presents an open-source workflow for reconstructing the volume of material removed during excavation, applied to the masonry foundation levels of an Iron Age structure at Ca' Nova di Albareto (Parma). The workflow is carried out entirely in CloudCompare on pairs of georeferenced point clouds acquired with a low-cost LiDAR-equipped iPad at successive stages of the excavation. The approach relies on the principle that the difference between two consecutive scans corresponds to the excavated material. The cloud-to-cloud distance between the two scans, stored as a scalar field, highlights the removed masonry, which is isolated with a threshold. Its boundary, extracted as an envelope, then defines the area clipped from the later scan to obtain the surface beneath. Both surfaces are merged and closed through interpolation, artificial and internal points are removed with the Hidden Point Removal plugin, and a Poisson surface reconstruction produces a closed mesh. The result is a 3D model of each removed foundation level, useful both for visualization and for calculating the volume removed. Repeated for each recording interval, the procedure produces a sequence of meshes that reconstructs the excavation process step by step. The workflow tested at Ca' Nova di Albareto can be applied to other excavation contexts and offers a quantitative basis for studying stratigraphic relationships and site formation processes.